Girls Build Bots — Brownie Digital Game Design

Badge-aligned workshop guide for robotics-team-led outreach

Level: Brownies (Grades 2–3) · **Badge:** Coding for Good — Digital Game Design **Series:** Coding for Good (Coding Basics · Digital Game Design · App Development) **Format:** One 75–90 minute session · **Ratio:** 1 mentor per 3–4 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on free browser tools and craft supplies the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Brownie Coding for Good badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: coding *for good*

Every badge in this series ties coding to helping people. Brownies should leave believing that code isn't just for games — it's a tool they can use to make something better for someone else. Mentors: name this out loud. "We're not just coding — we're making something fun and fair for a real player."

Vocabulary: code, game, goal, rule, sprite, score, playtest, bug, debug.

Digital Game Design (75–90 minutes)

Badge theme: Learning how digital games are designed — goals, rules, and fairness — and planning and building a simple game.

Materials

- Laptops/tablets with **MakeCode Arcade** (arcade.makecode.com) or **Scratch** (scratch.mit.edu)
- "Game recipe" planning sheet: My game's goal / How you win / What makes it fun
- A familiar playground or board game to dissect as a group

Activity 1 — What Makes a Game a Game? (15 min)

Take apart a game everyone knows (tag, tic-tac-toe). Find its **goal**, its **rules**, and what makes it **fair and fun**. List them on the board — these are the same pieces every video game designer starts with.

Activity 2 — Design On Paper First (15 min)

Before touching a screen, each Scout fills a “game recipe” sheet: what’s the goal, how do you win, what makes it fun? Designers always plan before they build. Mentors help each Scout keep it *small* — one goal, one way to win.

Activity 3 — Build a Collector Game (40 min)

Mentor-led progressive build in MakeCode Arcade: - A player sprite that moves with the arrow keys - A “good thing” to collect (a star, an apple) - A score that goes up when you collect it - A win message when the score hits a target - Fast finishers: add a sound, or a “bad thing” to avoid

Because there’s a clear collect-and-score loop, every Scout finishes with a game she can actually play.

Activity 4 — Playtest Swap (10 min) + Wrap (5 min)

Scouts swap seats and play a neighbor’s game. One kind thing they liked, one idea to make it better (“two stars and a wish,” Brownie-sized). **Playtesting** is a real game-design job. Closing: hands up if your game changed after someone played it.

> **Robot moment (optional):** Share for 60 seconds how a brand-new game is invented for robotics competitions every single year — the team designs strategy around fresh rules, exactly like designing a game.

Completion Checklist — Digital Game Design

By the end of the session, to earn the badge, each Scout can: - [ ] Name the parts of a game: goal, rules, and what makes it fair/fun - [ ] Plan a simple game on paper before building it - [ ] Build a playable game with a goal and a way to win - [ ] Improve her game after someone else playtests it

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Mentor Guide (for the robotics team)

- **You don’t need a robot for this.** It runs entirely on free browser tools and paper. If you brought a robot, the “robot moment” box is a 60-second add-on — not the main event.
- **Never touch the Scout’s project.** Hands behind your back; guide with questions.
- **Celebrate bugs out loud** — “you found a bug, that’s what real programmers do all day” — every single time.
- **Keep it small.** Brownies succeed when the game is one goal and one way to win. Rein in over-ambition gently.
- **Watch the clock.** Brownies fade after ~25 minutes on one thing; move to the next segment on schedule even if unfinished.
- **Name the “for good.”** A good game is fair and fun *for the player* — that’s designing with someone else in mind.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Brownie Coding for Good badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.